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PRD: AceFluency Spoken LMS Platform

0. Document Purpose

This Product Requirements Document (PRD) serves as the single source of truth for the product requirements, user experiences, and functional boundaries of the **AceFluency LMS Platform** expansion. It is formulated specifically for Product Managers, UX Designers, Frontend and Backend Architects, and QA Engineers.

All terminology inside this document is anchored to the strict definitions established in the **Glossary (§3)**; the use of synonyms is prohibited to avoid alignment drift. The core product requirements are divided into stable **Functional Requirements (§4)** containing explicit actors, capabilities, and testable consequences to ensure ease of automated QA verification. Critical strategic logic is mapped to an **Assumptions Index (§9)** for explicit tracking and validation. Technical integration structures, database mappings, and API schemas are decoupled from this document and reside in the **Technical Addendum (`addendum.md`)** to preserve a strict functional-strategic focus in the main PRD.

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1. Vision

For aspiring professionals, graduates, and remote freelancers in developing corridors (specifically starting with Bangladesh-to-GCC economic zones), spoken English is the highest-leverage economic gateway. Yet, four out of five bilingual learners face massive emotional barriers, pronunciation intimidation, and speech hesitation due to fear of judgement.

AceFluency is an accent-positive, 100% immersive English-only speaking playground. Rather than offering passive grammar rules, AceFluency targets the expansion of a user's cumulative **Speaking Hours** through engaging, ultra-low-bandwidth, scenario-driven voice circles called **Digital Adda**. By transforming spoken language practice from a solitary chore into a low-pressure, high-frequency social habit, AceFluency enables users to build natural, communicative fluency that helps them secure remote freelance contracts, pass corporate interviews, and thrive in global digital services.

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2. Target User

2.1 Jobs To Be Done (JTBD)

- **Functional (Remote Freelancer):** I need to verbally negotiate scope, timelines, and payment rates directly with global clients on contract marketplaces (Upwork, Fiverr) without stuttering or depending on text translations.

- **Social (Corporate Job-Seeker):** I want to participate confidently in bilingual or English-only panel interviews, showing high professionalism and technical articulation to senior managers without grammatical hesitation.
- **Emotional (Anxious Learner):** I need a safe, non-judgmental playground where I can make speaking mistakes, mispronounce technical words, and struggle with sentence structure without being criticized or mocked by peers.
- **Contextual (Low-Bandwidth Mobile User):** I need to practice daily spoken English while commuting on erratic mobile data networks (2G/3G) using a low-cost Android phone, without exhausting my daily mobile internet budget.

2.2 Non-Users (v1)

- **Native-Accent Pursuers:** Individuals seeking to eliminate their natural regional accents or mimic native British/American phonetics. v1 is explicitly focused on *communicative clarity* and *conversational confidence*, not accent reduction.
- **Academic Exam candidates:** Students looking to cram theoretical structures for IELTS/TOEFL written sections. AceFluency optimizes for real-time conversational fluency.

2.3 Key User Journeys

The platform utilizes three globally numbered **User Journeys (UJ)** that serve as direct baselines for downstream UX design and engineering validation.

- **UJ-1. Fahim completes his first conversation before registration.**
    - **Persona + context:** Fahim is a 22-year-old freelance web developer in Barisal, operating on a low-end Android device with a highly restricted 3G data plan.
    - **Entry state:** Unauthenticated, guest profile mode, landing screen contains a single "Join Speaking Circle" action block.
    - **Path:** Fahim taps "Join Speaking Circle". The system requires no sign-up; it prompts him to pick the "Freelance Client Negotiation" scenario. He taps 'Match'. Within 2.5 seconds, he is bridged into an active, audio-only group call with two other peer learners.
    - **Climax:** Fahim reads a contextual on-device visual speaking cue to introduce himself. He speaks for 3 minutes, receiving real-time, low-latency audio from peers without call drops, successfully completing a 15-minute mock negotiation.
    - **Resolution:** As the call successfully terminates, Fahim gets a prompt showing he accumulated 15 **Speaking Hours** fractional minutes, offering to persist his progress by creating a free account.
    - **Edge Case:** If the matchmaking queue exceeds 2.5 seconds due to low concurrency, the system seamlessly inserts a highly responsive Conversational AI Agent into the slot, ensuring Fahim gets immediate speaking value.
- **UJ-2. Anika pays for a Micropayment Pass during a power cut.**

- **Persona + context:** Anika is a corporate applicant in Dhaka whose daily complimentary 15 minutes of speaking time have expired, and she must practice immediately before an interview.
- **Entry state:** Authenticated, premium tier is "free", active daily minutes remaining is equal to 0.
- **Path:** Anika taps "Instant Speaking Pass". She selects "MFS Micropayment Pass (3-Hour Access)" priced at 50 BDT. She is redirected to a localized SSLCommerz payment page.
- **Climax:** She enters her bKash mobile wallet PIN, completes the OTP submission, and receives a success text message. The payment gateway returns an instant asynchronous confirmation.
- **Resolution:** Anika is returned to the app within 120ms of payment completion, where her account is upgraded to "Active Premium", and she immediately boots a speaking session.

- **UJ-3. Rakib recovers from a vocabulary freeze.**
  - **Persona + context:** Rakib is an entry-level IT support executive in Chittagong who freezes during a practice scenario when asked to describe a technical system bottleneck.
  - **Entry state:** Active call session inside an audio-only digital circle room.
  - **Path:** Rakib stops speaking for 6 consecutive seconds. The system detects the prolonged silence while it is his active speaking turn.
  - **Climax:** An on-device **Phrase Whisperer** overlay slides up in under 500ms, displaying three custom, contextual phrases ("The primary bottleneck is related to...", "We are currently observing latency because...", "To mitigate this drop, we should...").
  - **Resolution:** Rakib reads one of the visual phrases aloud, breaking his hesitation and resuming smooth verbal communication.

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3. Glossary

Downstream workflows, design specs, and automated test libraries must use these terms exactly as defined below. No synonyms or alternate phrasing is allowed.

- **Digital Adda:** A low-latency, audio-only group conversation room supporting 3 to 5 concurrent participants configured on a self-hosted Selective Forwarding Unit (SFU) network.
- **Speaking Hours:** The canonical metric tracking a user's accumulated, real-time verbal engagement duration inside active Digital Adda sessions, measured strictly in seconds and displayed to the user in formatted minutes/hours.
- **ELO Rating Score:** A dynamic mathematical value ranging from 400 to 2400 (defaulting to

1000) that models a user's verbal English proficiency, updated dynamically based on session participation and peers' feedback.

- **Micropayment Pass:** A time-bound premium checkout ticket (e.g., 50 BDT for 3 hours of unlimited room access) integrated through localized mobile financial services.
- **Phrase Whisperer:** An on-device, lightweight contextual text prompt engine that detects conversational pauses on the client device and displays interactive vocabulary suggestions.
- **Selective Forwarding Unit (SFU):** A low-level WebRTC server architecture that forwards audio media streams dynamically between participants without transcoding, minimizing CPU load and bandwidth.
- **Comfort Noise Generation (CNG):** A technical media optimization that generates standard synthesized background hums on the client device during speech pauses, working alongside Discontinuous Transmission (DTX) to save cellular packets.
- **SSLCommerz:** The primary domestic payment gateway integrated to capture local mobile wallets (bKash, Nagad, Rocket) and cards inside Bangladesh.
- **Conversational AI Agent:** An automated speech synthetically-driven participant spawned as a placeholder cohort in matchmaking sessions when peer density falls below active thresholds.

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4. Features

4.1 Low-Bandwidth Voice Rooms (Digital Adda)

Description: Digital Adda provides low-latency speaking circles that operate efficiently over highly degraded 2G/3G mobile networks. It forces the complete rejection of video streams to protect data limits, allowing users in remote locations with up to 35% packet loss to speak smoothly. It uses the OPUS codec with adaptive bitrate downscaling (6-8 kbps), dynamic suppression of ambient noise, and comfort noise overlays to preserve a immersive group vibe whilst keeping cellular consumption under 5MB per 30 minutes.

Functional Requirements:

FR-1: Audio-Only Ingestion Gate

The Selective Forwarding Unit (SFU) must filter and drop any incoming video track publications at the socket boundary. Realizes UJ-1.

- **Consequences (testable):** The room session instantly rejects any mobile WebRTC client publishing a track other than `audio` with a clean `WebRTCTrackRejectedException`.
- **Out of Scope:** Standalone user avatars uploading or real-time facial expressions rendering during active voice connections.

FR-2: Under-Silence Bandwidth Halting

During periods of silence where the active speaker's voice falls below -45dB, the WebRTC client must halt processing audio packets. Realizes UJ-1.

- **Consequences (testable):** Discontinuous Transmission (DTX) is activated; network telemetry packets drop to exactly zero per second during a speaker's silence.
- **Consequences (testable):** The target device's speaker plays a local synthesized ambient hum via Comfort Noise Generation (CNG) to prevent the user from perceiving a dropped line.

FR-3: Packet Loss Recovery

The audio streaming layer must dynamically decrease sample latency and use Forward Error Correction (FEC) when packet loss rises. Realizes UJ-1.

- **Consequences (testable):** Under 30% synthetic packet loss injected on a 3G network link, the audio intelligibility score (POLQA/MOS) must remain above 3.2, and audio playback must not crackle or drop.

Feature-specific NFRs:

- **In-Room Audio Latency:** The round-trip delay between any two client devices (measured from speaker A speaking to speaker B hearing) must be less than 50ms over standard 3G cell connections.

4.2 Chess-Inspired Skill Matchmaking

Description: Matchmaking connects active users into Digital Adda rooms containing 3 to 5 participants. It aims to avoid peer intimidation by keeping all matched cohorts within a strict proficiency boundary (maximum 100 ELO Rating Score points difference). It manages queues asynchronously in cache rather than reading/writing SQL tables, ensuring matches conclude in under 2.5 seconds. If a match cannot be fulfilled quickly during off-peak periods, the queue automatically triggers a Conversational AI Agent to step in as a peer.

Functional Requirements:

FR-1: ELO Proficiency Grouping

The system must group queue applicants into rooms according to their ELO Rating Score. Realizes UJ-1.

- **Consequences (testable):** Under competitive peak concurrency, the absolute difference in ELO rating between the highest-skilled participant and the lowest-skilled participant in any generated room must not exceed 100 points.
- **Consequences (testable):** Under lower queue concurrency, the ELO search window widens by +100 ELO points for every 3.0 seconds spent in the queue, capped at a maximum ELO disparity of 300 points, to prevent excessive matchmaking latency.

FR-2: Asynchronous Matching Cache

Wait-queues must reside entirely in high-speed, volatile cache blocks partitioned by scenario categories, completely isolated from SQL transaction engines. Realizes UJ-1.

- **Consequences (testable):** The server-side match evaluation loop must complete inside Redis in under 5ms, avoiding write locks on standard postgres transaction tables.

FR-3: AI Warm-up Cohort Injection

When a queue applicant stays unmatched inside a specific scenario category for more than 2.5 seconds, the matchmaking queue engine must generate an active room populated with Conversational AI Agents. Realizes UJ-1.

- **Consequences (testable):** The guest or user is transitioned out of the queue and bridged into the room setup immediately on the 2.6-second tick, receiving instant spoken prompts from the automated cohort.

Feature-specific NFRs:

- **Queue Completion Lifecycle:** The absolute duration between a user tapping "Match" and the client launching a LiveKit room subscription must be less than 2.5 seconds under standard peak loads.

4.3 Localized Micropayment & Subscriptions

Description: This feature governs early access hooks and monetization layers inside the LMS platform. To minimize friction, the registration gate is deferred completely (*Zero-Log* setup); guests can consume their initial 15-minute practice session and matching room directly, with sign-up prompts triggering only at the emotional peak of call termination. When daily free minutes are spent, users upgrade immediately through a domestic checkout page integrated with SSLCommerz, converting local mobile wallets (bKash, Nagad) into instant Active Premium speaking limits in under 120ms.

Functional Requirements:

FR-1: Deferred Auth Gate (Zero-Log)

The guest speaking circle initialization must operate without requiring an authenticated user session or token registration. Realizes UJ-1.

- **Consequences (testable):** Unregistered visitors can execute the full matching sequence and speak in a complete 15-minute room session without creating a row in the secure identity table.
- **Consequences (testable):** The client stores a transient device token to persist anonymous Speaking Hours metrics locally in device cache.
- **Consequences (testable):** The anonymous local cache is evicted and local speaking metrics reset if the guest account remains unassociated with a secure registered credential for more than 7 running calendar days.

FR-2: Localized Payment Integration

The system must support bKash, Nagad, and local Bangladeshi debit and credit card systems through a unified domestic checkout bridge. Realizes UJ-2.

- **Consequences (testable):** Tapping payment redirects the mobile viewport to a localized gateway environment under HTTPS, supporting secure local currencies (BDT).

FR-3: Instant Subscription Upgrade

The system must parse payment gateway success callbacks asynchronously and refresh the user's entitlements in under 120ms. Realizes UJ-2.

- **Consequences (testable):** Upon receiving a valid SSLCommerz payment callback, the backend asynchronously issues a token update such that the client's premium speaks status transitions to "Active Premium" in under 120ms.

4.4 Contextual "Phrase Whisperer"

Description: The Phrase Whisperer acts as a real-time conversational safety net. If a participant stops speaking for 6 consecutive seconds (due to vocabulary freeze or anxiety), the client app immediately pops up contextual, scenario-specific prompt hints. To prevent vocal hitching, the engine executes 100% locally on the user's mobile device, rendering overlays in under 500ms without reaching out to slow network cloud APIs.

Functional Requirements:

FR-1: Local Voice Activity Pause Detection

The client device's audio input processor must track the user's speaking volume threshold and trigger an alert if a pause is detected in their active turn. Realizes UJ-3.

- **Consequences (testable):** If the input audio volume dips below -45dB and remains there for exactly 6.0 seconds while the room token denotes the user's speaking slot, a local signal is dispatched.

FR-2: On-Device Contextual Phrase Overlay

The application must store a local dictionary of scenario prompts and display relevant visual speech overlays instantly. Realizes UJ-3.

- **Consequences (testable):** The app renders a translucent contextual display containing three relevant visual sentence starters matching the active scenario category in under 500ms from the local pause signal.
- **Consequences (testable):** The rendering logic executes with 0ms network round-trip overhead, operating completely offline if the cellular data link temporarily breaks.

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5. Non-Goals (Explicit)

To prevent scopes creep and keep engineering focused on core delivery priorities, the following non-goals are formally declared:

- **Bilingual Bangla/English UI:** The mobile and web application interface language is strictly 100% English. Providing local translation interfaces breaks target-language baseline immersion, which is a core educational constraint. [Strategic Override: Approved 2026-05-28]

- **Native Accent Modification:** Standardizing pronunciation coaching on American/British accents is out of scope. Lessons and feedback focus exclusively on *International English Communicative Clarity*.
- **Pre-Session User Verification:** The system will not perform KYC or phone authentication checkouts prior to a guest's first 15-minute speaking session.
- **Group Video Streaming:** The system will never initialize camera, video tracks, or screen-sharing capabilities inside any Digital Adda room.

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6. MVP Scope

6.1 In Scope

- Self-hosted low-latency audio Selective Forwarding Unit (SFU) handling 3-5 person speaker rooms.
- Biome-formatted shared react-native client app structures.
- Redis ZSET matchmaking engine matching users within 100 ELO score bands.
- Contextual Phrase Whisperer local overlay dictionaries for the initial three career scenarios:
    1. *Freelance Contract Negotiation*
    2. *Software Architecture Peer Review*
    3. *Corporate Project Status Update*
- 100% English-only mobile interface with Zero-Log onboarding (deferred sign-up up to 15 minutes of speaking duration).
- Local payment processing through SSLCommerz support for bKash and Nagad mobile finance.

6.2 Out of Scope for MVP

- **Pre-Session Audio Quality Testing:** Dynamic server evaluation of microphone hardware quality before matching. [Revisit in v2]
- **Automatic Peer Moderation:** Automated transcription-based toxicity auditing or direct voice bans on bad actors. [Revisit in v2]
- **Custom Scenario Builders:** Allowing corporate partners to create localized custom career scenarios from their own admin templates. [Revisit in v3]

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7. Success Metrics

The following metrics are used to validate the product's strategic success and must be monitored continuously post-production.

Primary Metrics

- **SM-1: Daily Speaking Hours Volume:** The cumulative sum of speaking durations collected across all Digital Adda rooms globally per day. Target: > 50,000 Spoken Hours per week. Validates FR-1, FR-3.
- **SM-2: Dynamic Match Completion Rate:** The proportion of queue initialization triggers that successfully bridge into active LiveKit room sessions within 2.5 seconds. Target: > 96%. Validates FR-2.

Secondary Metrics

- **SM-3: MFS Checkout Conversion Rate:** The percentage of users starting a 50 BDT micropayment flow that secure a successful payment return inside SSLCommerz. Target: > 82%. Validates FR-2.

Counter-Metrics (Do Not Optimize)

- **SM-C1: Video track activation count:** Must remain exactly 0. Any attempt to introduce video components to optimize user retention will directly compromise low-bandwidth targets (SM-1 data payload) and must be blocked.
- **SM-C2: Direct Sign-up Conversion Rate:** While we measure registration, we must not optimize this metric by moving the sign-up gate *before* the first speaking call, as doing so would directly suppress early matchmaking pool density.

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8. Open Questions

1. **Matchmaking Pool Density Balancing:** How will the ELO rating criteria adjust dynamically during off-peak windows if the concurrency drops below critical mass and AI agents are disabled?
    - *Hypothesis to test:* Introduce an adaptive ELO search window that widens by +100 ELO points for every 3 seconds spent in the queue, capped at +300 ELO total disparity.
2. **Attribution Loss on Deferred Signup:** Will anonymous attribution metrics remain accurate across modern mobile browser cookie limits when users delay their signup by 15 minutes?
    - *Audit check:* Implement local persistent analytics logs tracking the initial acquisition source ID prior to displaying the first call setup.

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9. Assumptions Index

The following assumptions represent structural strategic choices made in this specification and must be explicitly validated during early testing stages:

- **[ASSUMPTION-1: English-Only Friction]:** Assumes that a 100% English-only mobile application user interface will not cause premature guest bounce, and that the onboarding flow is expressive enough to guide non-speakers using universal visual icons.

- **[ASSUMPTION-2: Data Budget Adequacy]:** Assumes that rural Bangladeshi cellular network profiles (specifically 2G/3G in Barisal/Sylhet) can support steady OPUS-compressed WebRTC audio links under 35% packet loss without exceeding a 5MB per 30 minutes user budget limit.
- **[ASSUMPTION-3: Micropayment Model Habituation]:** Assumes that low-income remote freelancers in Bangladesh are comfortable paying small incremental fees via bKash (e.g., 50 BDT) for time-bound professional practicing access, rather than expecting a purely ad-supported subscription standard.